

Overall Performance Comparison (Mean $\pm$ 95% CI)

	Total Energy (kJ) ↓	High-Priority Satisfaction (%) ↑	Medium-Priority Satisfaction (%) ↑	Low-Priority Satisfaction (%) ↑
2D-AUTO (46–50 m)	81.5 $\pm$ 0.5	0.6 $\pm$ 0.9	61.0 $\pm$ 5.1	66.7 $\pm$ 4.2
Two-Stage (decoupled)	79.4 $\pm$ 1.0 ***	54.9 $\pm$ 5.7 ***	66.8 $\pm$ 5.2 n.s.	52.7 $\pm$ 5.4 ***
Coupled-Greedy (ablation)	80.1 $\pm$ 1.2 *	54.9 $\pm$ 5.6 ***	71.5 $\pm$ 5.1 **	64.9 $\pm$ 6.1 n.s.
Strong-Coupled (deterministic)	78.5 $\pm$ 1.5 ***	70.2 $\pm$ 7.3 ***	80.8 $\pm$ 3.8 ***	74.2 $\pm$ 6.2 *

Bold marks the best value per column.

*Markers: paired t-test vs 2D-AUTO — *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant.*

$n = 20$ city seeds; $\pm$ is the Student-t 95% CI.

Ours vs Two-Stage (decoupled), paired: energy $p = 0.332$, high $p = 0.001$, medium $p = 0.001$, low $p < 0.001$.

Means match Figs. 5-6; ↓ lower is better, ↑ higher is better.